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17 LAMBERT. IRRATIONALITY OF π

By 1750 the number π had been expressed by infinite series, infinite products, and infinite continued fractions, its value had been computed by infinite series to 127 places of decimals (see Selection V.15), and it had been given its present symbol. All these efforts, however, had not contributed to the solution of the ancient problem of the quadrature of the circle; the question whether a circle whose area is equal to that of a given square can be constructed with the sole use of straightedge and compass remained unanswered. It was Euler's discovery of the relation between trigonometric and exponential functions that eventually led to an answer. The first step was made by J. H. Lambert, when, in 1766–1767, he used Euler's work to prove the irrationality not only of π, but also of e.

Johann Heinrich Lambert (1728–1777) was a Swiss from Mülhausen (then in Switzerland). Called to Berlin by Frederick the Great, he became a member of the Berlin Academy and thus a colleague of Euler and Lagrange. His name is also connected with the introduction of hyperbolic functions (1770), with perspective (1759, 1774), and with the so-called Lambert projection in cartography (1772).

Lambert published his proof of the irrationality of π in his "Vorläufige Kenntnisse für die, so die Quadratur und Rectification des Circuls suchen," *Beyträge zum Gebrauche der Mathematik und deren Anwendung* 2 (Berlin, 1770), 140–169, written in 1766, and in more detail in the "Mémoire sur quelques propriétés remarquables des quantités transcendentes circulaires et logarithmiques," *Histoire de l'Académie, Berlin, 1761* (1768), 265–322, presented in 1767. They have been reprinted in the *Opera mathematica*, ed. A. Speiser (2 vols.; Füssli, Zurich, 1946, 1948), I, 194–212, II, 112–159. The following text is a translation from pp. 132–138 of vol. II. Lambert writes tang where we write tan. See also F. Rudio, *Archimedes, Huygens, Lambert, Legendre. Vier Abhandlungen über die Kreismessung* (Teubner, Leipzig, 1892).

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37. *Now I say that this tangent* [$\tan \varphi/\omega$] *will never be commensurable to the radius, whatever the integers* ω, φ *may be.*[1]

[1] In the previous sections Lambert expands $\tan v$, v an arbitrary arc of a circle of radius 1, into a continued fraction, and gets for $v = 1/w$

$$\tan v = \cfrac{1}{w - \cfrac{1}{3w - \cfrac{1}{5w - 1}}} \quad \text{etc.}$$

Investigating the partial fractions and their residues, he finds infinite series like

$$\tan v = \frac{1}{w} + \frac{1}{w(3w^2 - 1)} + \frac{1}{(3w^2 - 1)(15w^3 - 6w)} + \cdots$$

and shows (in §34) that these series converge more rapidly than any decreasing geometric series. Then, if $w = \omega : \varphi$, ω, φ being relatively prime integers, he finds for the partial fractions of $\tan v$ (§36):

$$\frac{\varphi}{\omega}, \quad \frac{3\omega\varphi}{3\omega^2 - \varphi^2}, \quad \frac{15\omega^2\varphi - \varphi^3}{15\omega^3 - 6\varphi^2\omega}, \quad \frac{105\omega^3\varphi - 10\omega\varphi^3}{105\omega^4 - 45\omega^2\varphi^2 + \varphi^4}, \quad \text{etc.,}$$

and (§37):

$$\tan \frac{\varphi}{\omega} = \frac{\varphi}{\omega} + \frac{\varphi^3}{\omega(3\omega^2 - \varphi^2)} + \frac{\varphi^5}{(3\omega^2 - \varphi^2)(15\omega^3 - 6\omega^2\varphi)} + \text{etc.}$$

Then follows the text which we reproduce.